ACME A2 REPORT

Group number: 09

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Practical Network Defense - Assignment 2

Initial brainstorming

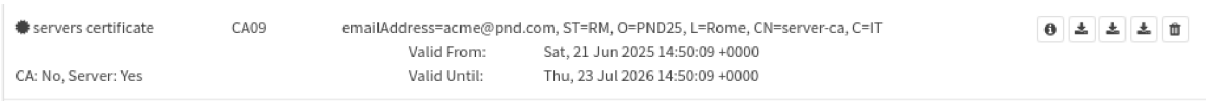
We have to configure an OpenVPN between Road warriors and the Main firewall and an IPSec VPN between the main router firewall and the internal router firewall.

We need to understand how to make a connection between a client from the outside (that is, a Road warrior employee), and the Main router firewall. Then, we need to understand how to make an IPSec tunnel between the two router firewalls.

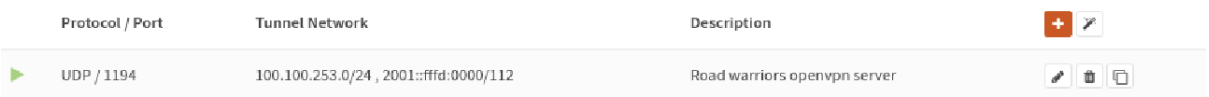
OpenVPN will rely on a PKI, while IPSec will be implemented through a VTI (Virtual Tunnel Interface) and a route based setup, this because a VTI gives us a more granular control on how the traffic is routed among the different networks while passing through the IPSec tunnel between the two firewall-routers.

OpenVPN setup

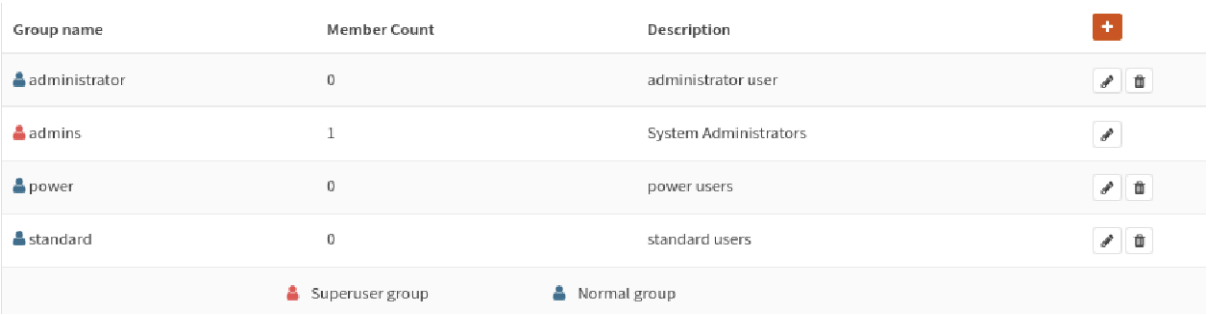
We first created in the Main firewall-router the Certificate Authority under System -> Trust -> Authorities and System -> Trust -> Certificate.



We then created the OpenVPN going in VPN -> OpenVPN -> Servers.



We created the groups related to the employees: System -> Access -> Groups. One for standard, one for power, and one for administrator users.



After that, we created the users that will use the VPN: Alice (standard user), Bob (power user), and Christina (administrator). To do that, we go into System -> Access -> Users and create three users with these username and password:

- alice: a=L?D44t6m'v?K6

- christina: +,g6cHL^dB_,v&9
- bob: }b6be#U#GR4TY

Also, other than the users, for each of them we created the certificates as shown:

alice	CA09	emailAddress=acme@pnd.com, ST=RM, O=PND25, L=Rome, CN=alice, C=IT Valid From: Sat, 21 Jun 2025 19:23:24 +0000 Valid Until: Thu, 23 Jul 2026 19:23:24 +0000	User Cert
CA: No, Server: No			
bob	CA09	emailAddress=acme@pnd.com, ST=RM, O=PND25, L=Rome, CN=bob, C=IT Valid From: Sat, 21 Jun 2025 19:26:01 +0000 Valid Until: Thu, 23 Jul 2026 19:26:01 +0000	User Cert
CA: No, Server: No			
christina	CA09	emailAddress=acme@pnd.com, ST=RM, O=PND25, L=Rome, CN=christina, C=IT Valid From: Sat, 21 Jun 2025 19:28:42 +0000 Valid Until: Thu, 23 Jul 2026 19:28:42 +0000	User Cert
CA: No, Server: No			

We set the server to “Remote Access (SSL/TLS)” in order to check for certificates without requiring user:password but instead using an authentication mechanism that relies on a Public Key Infrastructure (X.509). We then exported the .zip containing the file .p12 with client certificate, CA certificate and the private key, checking the .ovpn all was already set and then we tested the connection as shown:

```
└─$ sudo openvpn Road_warriors_openvpn_server_bob.ovpn
2025-06-22 16:38:32 WARNING: file 'Road_warriors_openvpn_server_bob.p12' is group or others accessible
2025-06-22 16:38:32 OpenVPN 2.6.12 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH
/PKTINFO] [AEAD] [DCO]
2025-06-22 16:38:32 library versions: OpenSSL 3.0.13 30 Jan 2024, LZO 2.10
2025-06-22 16:38:32 DCO version: N/A
2025-06-22 16:38:32 TCP/UDP: Preserving recently used remote address: [AF_INET]100.100.0.2:1194
2025-06-22 16:38:32 UDPv4 link local (bound): [AF_INET][undef]:0
2025-06-22 16:38:32 UDPv4 link remote: [AF_INET]100.100.0.2:1194
2025-06-22 16:38:33 [server-ca] Peer Connection Initiated with [AF_INET]100.100.0.2:1194
2025-06-22 16:38:33 sitnl_send: rtnl: generic error (-101): Network is unreachable
2025-06-22 16:38:33 TUN/TAP device tun0 opened
2025-06-22 16:38:33 net_iface_mtu_set: mtu 1500 for tun0
2025-06-22 16:38:33 net_iface_up: set tun0 up
2025-06-22 16:38:33 net_addr_ptp_v4_add: 100.100.253.10 peer 100.100.253.9 dev tun0
2025-06-22 16:38:33 net_iface_mtu_set: mtu 1500 for tun0
2025-06-22 16:38:33 net_iface_up: set tun0 up
2025-06-22 16:38:33 net_addr_v6_add: 2001::fffd:3/112 dev tun0
2025-06-22 16:38:33 Initialization Sequence Completed
```

As no specific constraints were outlined in the assignment requirements, we configured firewall rules to allow all types of traffic to the relevant destinations, based on the user's group membership.

	Protocol	Source	Port	Destination	Port	Gateway	Schedule	Description
								Automatically generated rules
								Floating rules
	IPv4+6 *	standard_users	*	DMZ net	*	*	*	Allow Standard user to access dmz
	IPv4+6 *	power_users	*	DMZ net	*	*	*	Allow power users to access dmz
	IPv4+6 *	power_users	*	INT_SERVERS	*	*	*	Allow power users to access internal servers
	IPv4+6 *	administrator_users	*	*	*	*	*	Allow all traffic from administrators

pass / pass (disabled) — block / block (disabled) — reject / reject (disabled) — log / log (disabled) — in / out — first match / last match

Active/Inactive Schedule (click to view/edit)

Alias (click to view/edit)

OpenVPN rules are evaluated on a first-match basis by default (i.e. the action of the first rule to match a packet will be executed). This means that if you use block rules, you will have to pay attention to the rule order. Everything that is not explicitly passed is blocked by default.

We verified the effectiveness of these rules by performing ping tests to the designated destinations, ensuring that traffic was correctly allowed based on group membership.

IPSec - Setup & Creation of the tunnel

First, we created three firewall rules both on the Main firewall-router (Internal interface) and the Internal firewall-router (External Interface) to allow for ESP, ISAKMP and NAT-T traffic, we did that in order to allow for exchange of IPSec traffic between the two router-firewall.

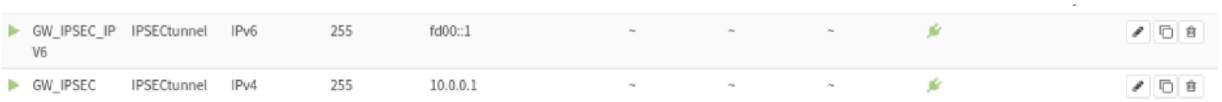


Then, we went to VPN -> IPSec -> Tunnel Settings, and we created Phase 1 and the two Phases 2 (IPv4 and IPv6) for IPSec for both the Main firewall-router and the Internal firewall-router. In main_fw -> vpn -> ipsec -> tunnel settings we took off the option install policy. This created conflicts with route-based tunnels. Then we set phase 2 mode to route-based, and as local and remote address set 10.0.0.1 and 10.0.0.2. These are the addresses that are used in the vti that the route-based mode creates where the IPsec tunnel resides. Main takes 10.0.0.1 and internal 10.0.0.2.

Do this in internal firewall too but specularly (invert source and local address)

Now the tunnel is set-up and we need to actually send packets into it. We need to create gateways and routes to accomplish this. In both firewalls create the gateways for 10.0.0.1 and 10.0.0.2 in system -> gateways -> configuration.

For the Internal Firewall:



For the Main Firewall:

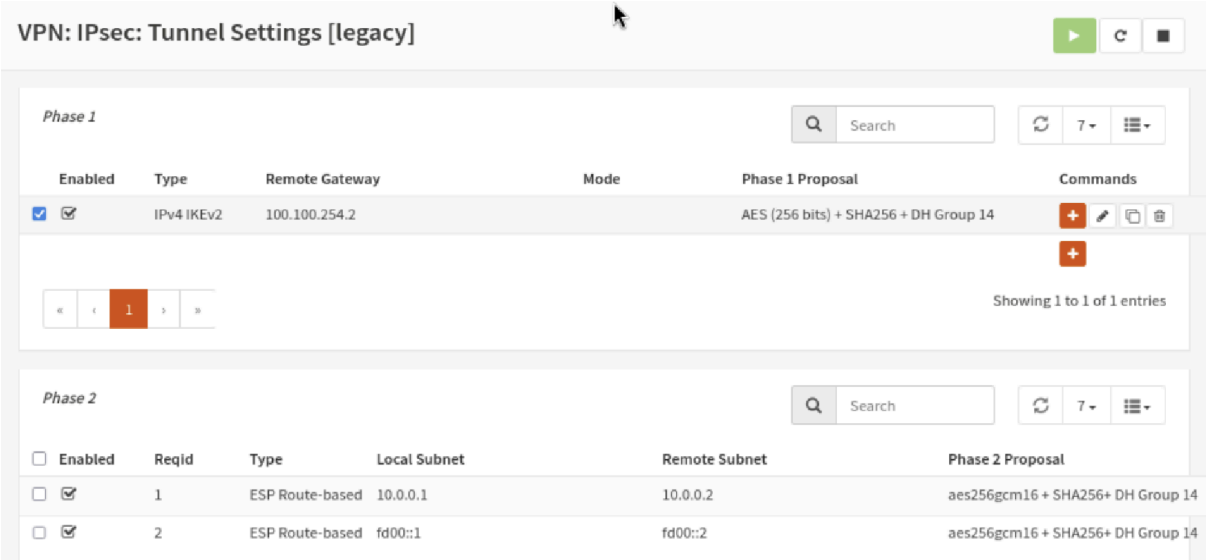


On main_fw modifies the two static_routes that existed. They sent packets directed to 100.100.1.0/24 and 100.100.2.0/24 to the internal_fw(100.100.254.2). We now need to send packets directed to these subnets to 10.0.0.2, so just modify the gateway to 10.0.0.2 and the interface to the vti.

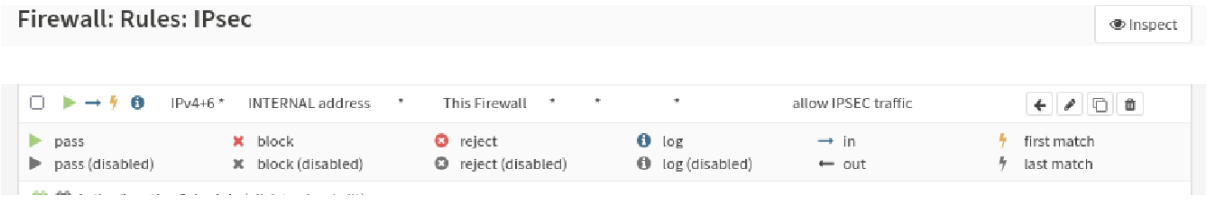
On internal_fw the default gateway is 100.100.254.1, so any packet that is not destined to the two local subnets will go through this. We need to override this. Create a static route for the network 0.0.0.0/0 and send it to gateway 10.0.0.1.

This accomplishes that the internal firewall checks in the routing table and does not find a match with the subnets, it looks in the static routes and always finds a match sending traffic in the tunnel. Before it would have found nothing in static routes and would have used the default gateway.

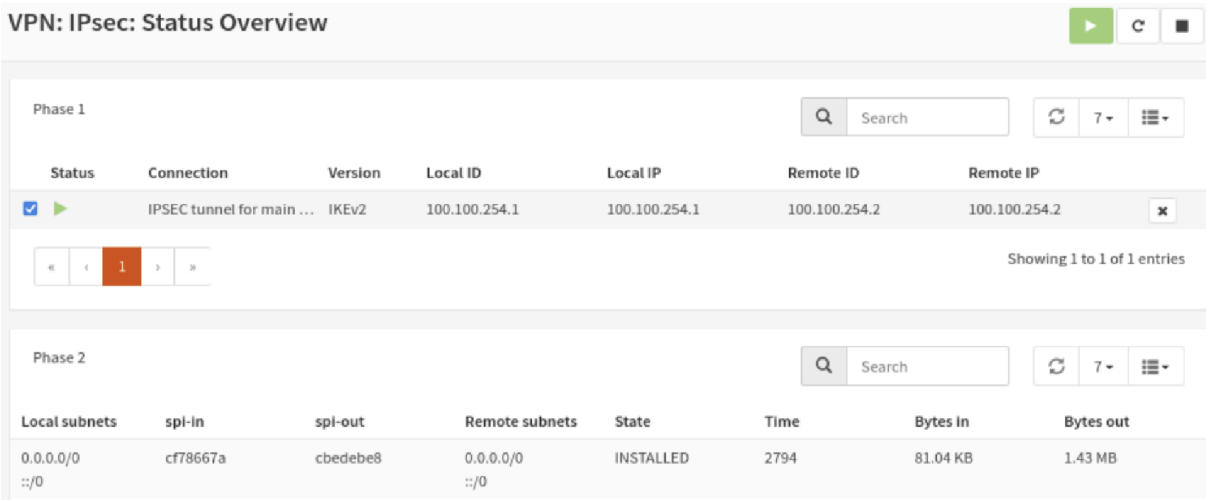
For ipv6 create another second phase with local addresses fd00::1 and fd00::2. Create routes and gateways for ipv6



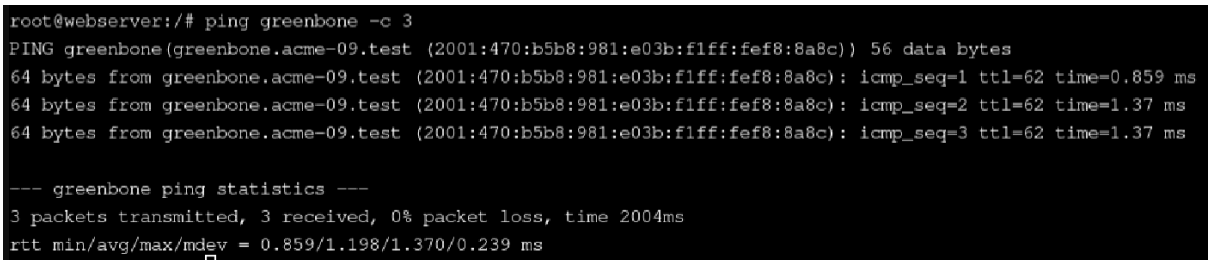
Now, in each firewall we set a rule to allow IPSec traffic. We went to Firewall -> Rules -> IPSec and added a new rule both for the Main firewall-router and for the Internal firewall-router.

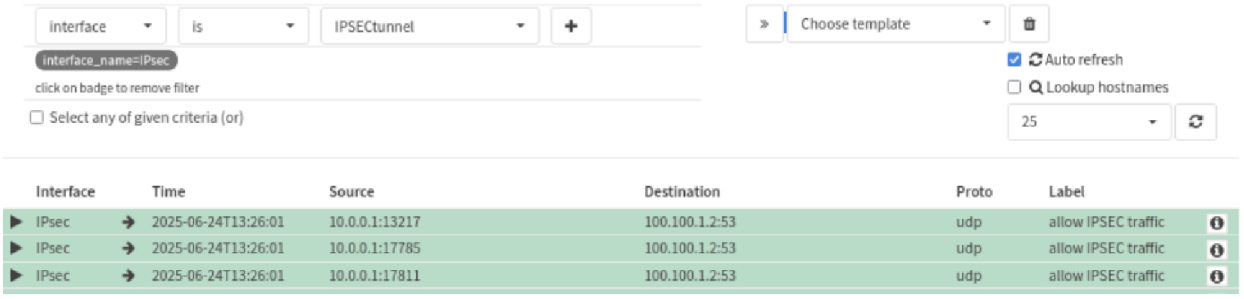


We finally check that the IPSec is up and running in VPN->IPSec->Status Overview.



Going on the Live View, we can clearly see that pinging from a host belonging to the DMZ (i.e. the web server) to a host in the internal servers network (i.e. greenbone) results in IPSec-related traffic being generated between the two router-firewalls. This confirms that the IPSec mechanism has been correctly implemented and it is functioning as expected, securing the communication as per the network policy.





High-Level Overview of ESP Route-Based IPsec Configuration

When using ESP route-based mode, OPNsense creates a Virtual Tunnel Interface (VTI) with its own IP addresses. Unlike policy-based VPNs, this VTI acts as a standard network interface. You then create specific static routes that direct desired traffic towards this VTI. This configuration ensures that only traffic explicitly routed through the VTI is encapsulated by IPsec, offering a clean separation of flows.

Final remarks

This assignment allowed us to learn how to use OPNsense and configure VPNs on it. Setting up the features took us quite a long time, as one of the main challenges we encountered was understanding how to correctly configure the tunnels along with the various IPSec options in OPNsense, with new functionalities often interfering with previously implemented ones. The available documentation and the built-in “full-help” mode were sometimes unclear/not detailed, which made it difficult to know which settings to enable or avoid. Moreover, we frequently had to reset the hosts and start the configuration from scratch. However, thanks to tools like Live View, we were able to monitor traffic effectively and validate our configuration. In addition, once we understood how to set everything up, the process became straightforward and easy to repeat in a short amount of time. Overall, the hands-on experience greatly enhanced our understanding of secure inter-zone routing using IPSec.